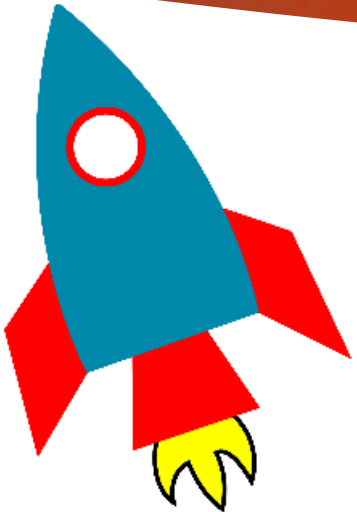


“

The Power of Inclusion

”

HOW INCLUSION CAN HELP EVERYONE IN THE FUTURE ON BOTH
WAYS



2018 – Dennie Declercq

NDC { London }

15-19 January 2018

“

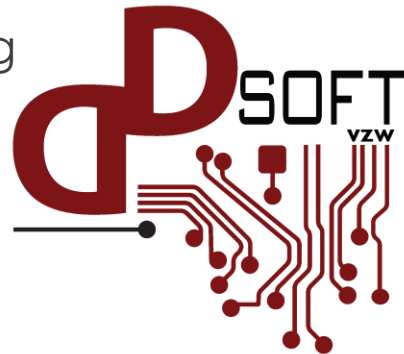
Introduction about myself

”

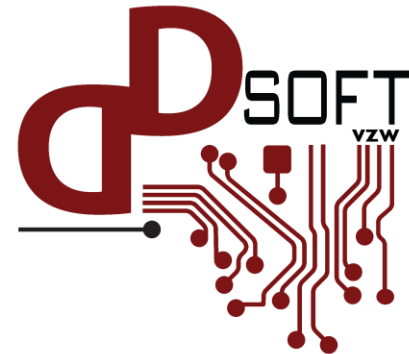
Who am I?



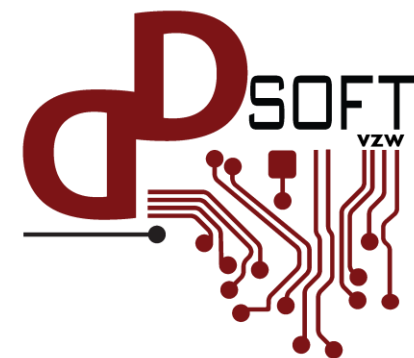
- ▶ Dennie Declercq
- ▶ Volunteer developer of apps, software and data solutions for people with a disability
- ▶ Representative and developer at DDSOFT vzw
- ▶ Inventor of #DreamingIsBelieving
- ▶ Positive thinker



The story of my life...



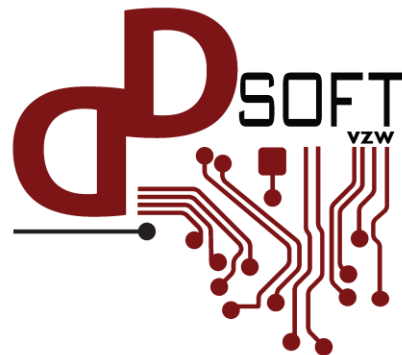
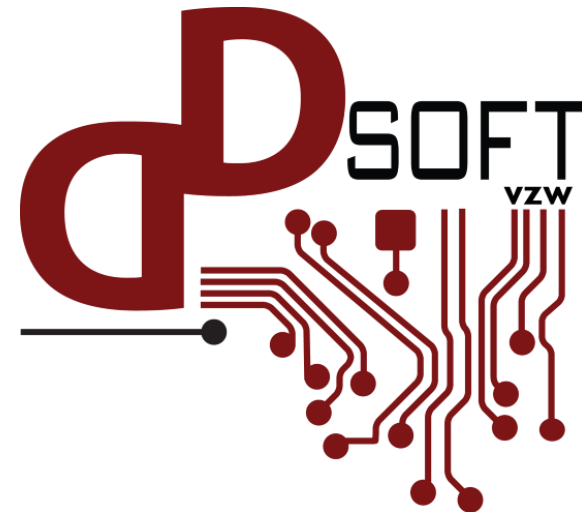
In my teenager years I was diagnosed
with autism



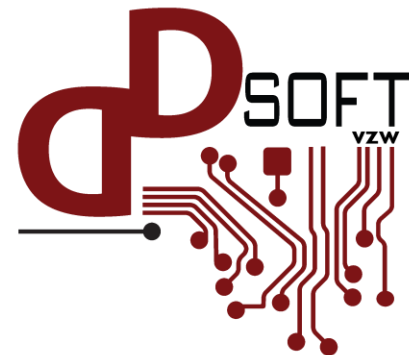
In a paid job I didn't fit...



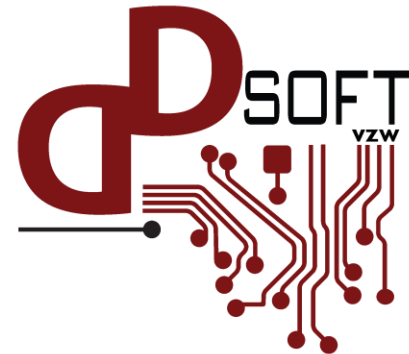
But I do what I like to do!



Continuous on my way



Bringing as much movement as possible



And Inclusion worked for me!



- ✓ I can Live on myself in “De Bezaan”
- ✓ I can work in the days on my DDSOFT project and in Ithaka
- ✓ I can live near to, no stress
- ✓ I cook by myself
- ✓ I go out to parties and conferences as much as I want
- ✓ I clean my house
- ✓ If I need a talk, I have good friends!

As you know, the proof of the pudding is in the eating!

Because the following inclusive environment is offered to me as person with autism, I really enjoy life!

As normal?

If I lived years ago the change was big that:

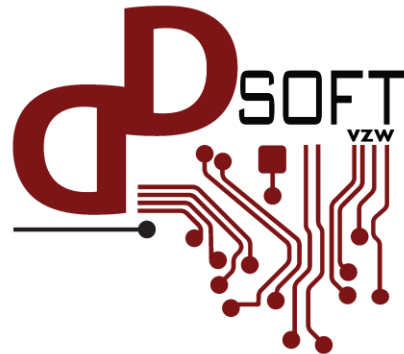
- I had to work in a special work, but live with stress
 - I had to live with my parents
 - I had to live in a cared environment
- With VERY LESS FREEDOM!!

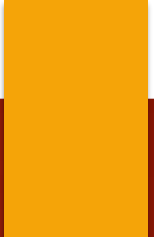


2017 was my year of
#DreamingIsBelieving



NDC { Sydney }





“

Let me take you on the journey
of inclusion

”

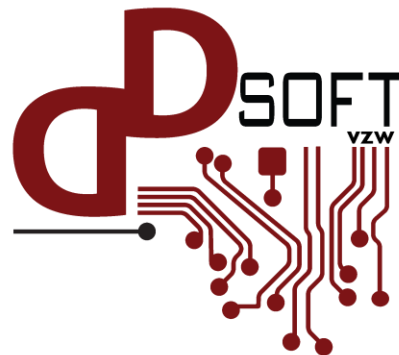
People with a disability



A Disability is **an impairment** that may be:

Physical
cognitive
Intellectual
Mental
Sensory

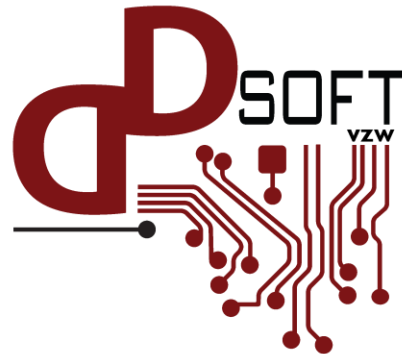
And combinations between those



Intellectual disability



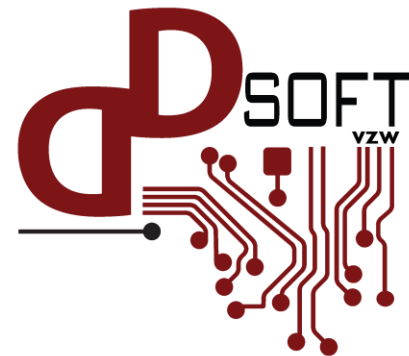
- ▶ Impaired intellectual functioning
- ▶ Impaired adaptive functioning



Visual impairments



- ▶ A **decreased ability to see**
- ▶ **Blindness** is used for complete or nearly complete vision-loss.



Autism

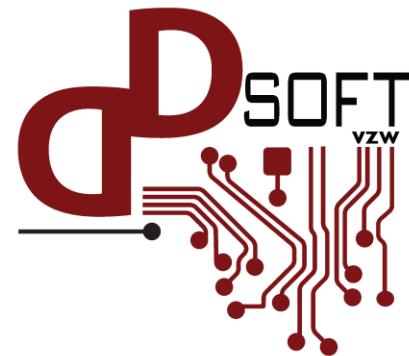
- ▶ A **neurodevelopmental disorder**
- ▶ Autism is **a spectrum!**
- ▶ There's **no one-way** approach



The early days



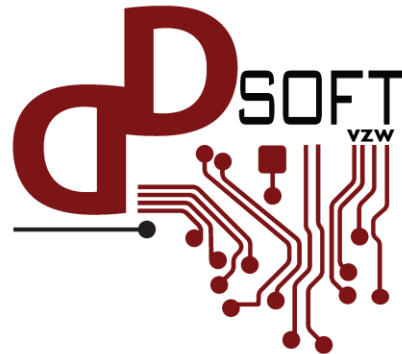
- ▶ **Taken away**
- ▶ There was **no place** in the social environment
- ▶ **Clinics** and **specialized institutions.**
- ▶ **Have to change their life** or have to be separated away...



The movement - Inclusion



- ▶ **Quality of life** (QoL) and **inclusion**. In short words, inclusion means: “people with a disability live and work together, in the same houses, the same environment as people without a disability, with **the needed adjustments par person**.”



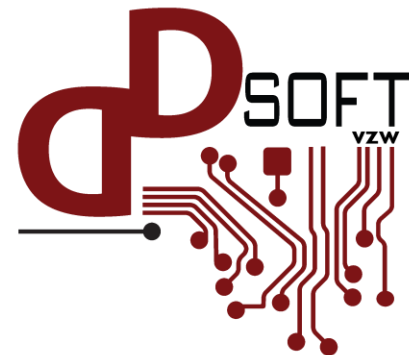
Schalock's domains of wellbeing.

“To have quality of life, each of the following aspects must score as good for people with a disability as without.”

Figure 2: Quality of life domains

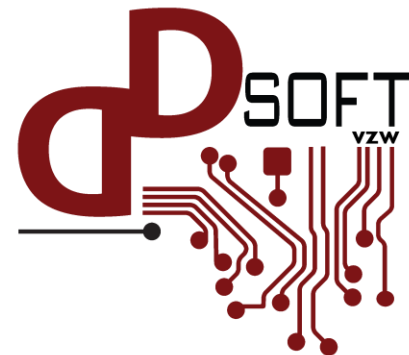
Interpersonal relations	Physical wellbeing	Self-determination	Social inclusion	Emotional wellbeing	Material wellbeing	Personal development	Rights
Affiliations	Health care	Choice	Natural supports	Safety	Ownership	Education and habilitation	Privacy
Affection	Mobility	Personal control	Integrated environments	Stable and predictable environments	Possessions	Purposive activities	Ownership
Friendships	Wellness	Decision	Participation	Positive feedback	Employment	Assistive technology	Due process
Intimacy	Nutrition	Personal goals	Barrier – free environments				
Interactions							

Source: Schalock, et al, 2002.



Van Gennep

- ▶ You can fill in your life yourself with your own normal, and special needs
- ▶ You can live, and do the same things as persons without a disability
- ▶ You are happy with the life you have



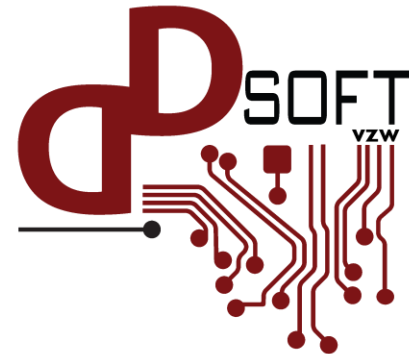
What inclusion really means

Real Inclusion

- People with disability doing the same activities as people without disability.
- People are living in the same district.
- People work together but the expectations are lower.

Not so real Inclusion

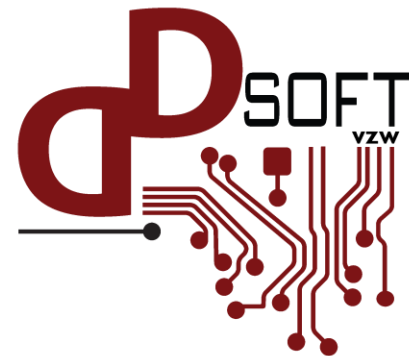
- A hostage for people with a disability.
- A school for kids with a disability.
- A working house specific for people with a disability.



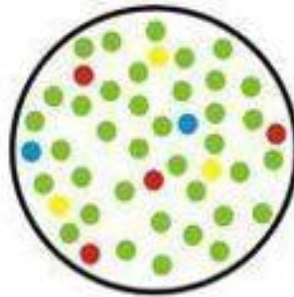
A way beyond diversion

“ DIVERSITY is being invited to the party. INCLUSION is being asked to dance.”

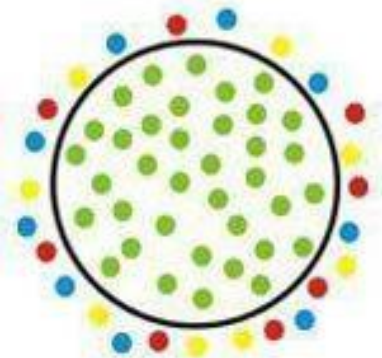
-Vernā Myers



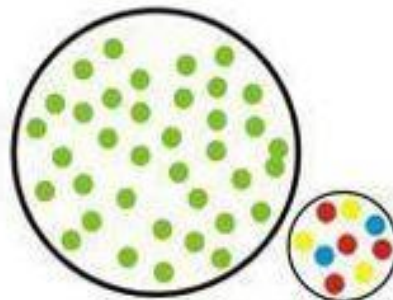
Not just integration



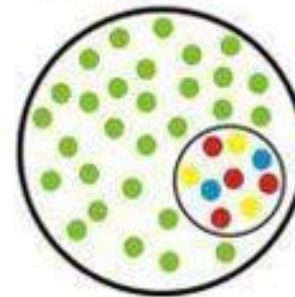
Inclusion



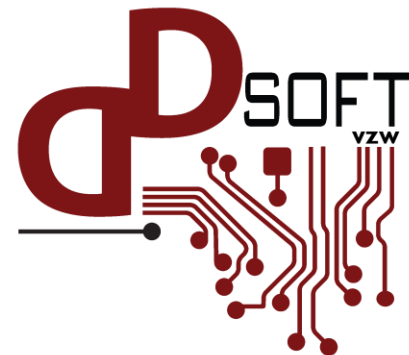
Exclusion



Segregation



Integration



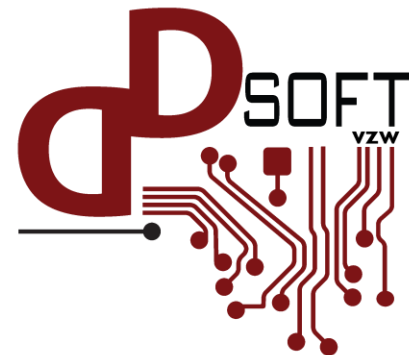
From caring to coaching

Caring

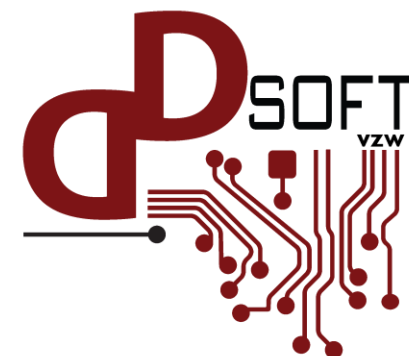
- Big institutions, clinics
- Professionals care people with disability
- Start point: “what is normal?”
- Many rules
- No in-between
- Big “you must take it all” packages
- Top (professional) – down (people with disability)

Coaching

- Small organizations
- Professional coaches the client
- Start point: “what wants the client?”
- Less rules
- A big “in between” zone
- Professional next to client



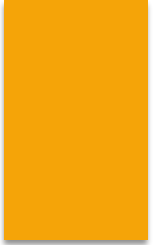
Inclusion in Flanders (Belgium) examples



“

Inclusive software and apps

”



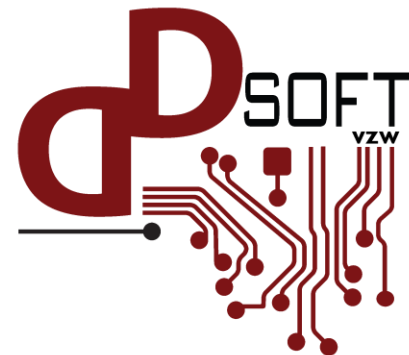
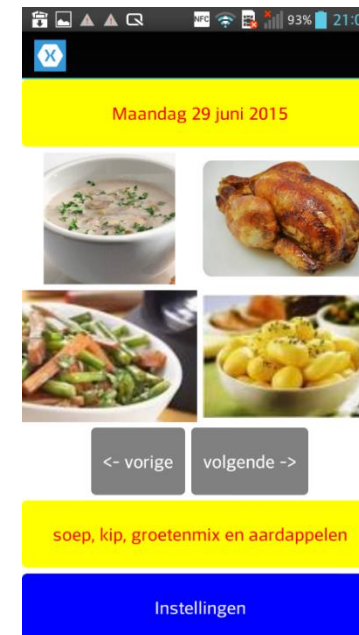
“If you want to create an inclusive environment, where people can live their own life, technology helps the people to find their way.”

Special apps and software, made for people with special needs can open many doors. You can give the world, packed in a simple software package.



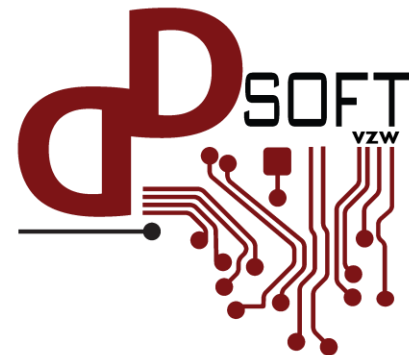
Basic rules:

- ▶ Simplify the layout
- ▶ Color by function
- ▶ Use as many pictures or pictographs as possible
- ▶ Add Text To Speech (TTS)
- ▶ Structured apps



Simplify the layout

- ▶ **Less** content per screen
- ▶ **Logical** content per screen
- ▶ **Light** or white **background**
- ▶ A few **accent** colors

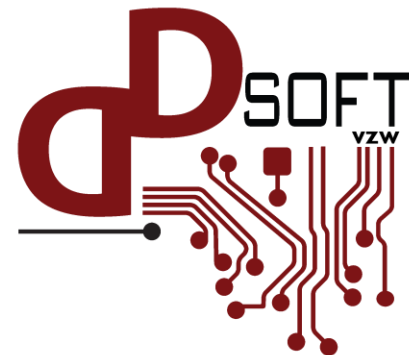


Color by function

I can speak !

I can navigate !

I open an other page!



Pictures and pictographs

► What means the following?

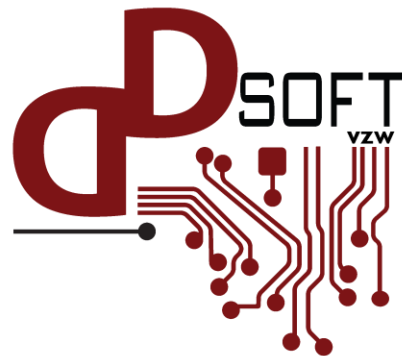
► **Kip**

► The answer is:



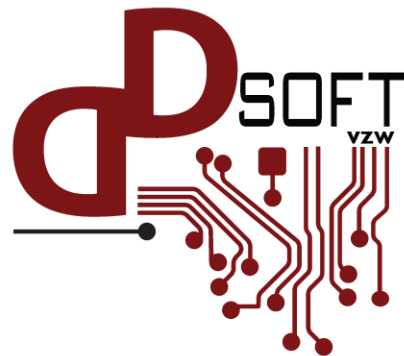
For some people with a disability, who cannot read, Letters and words means nothing.

An other language for us. Everybody (except people with visual impairments) can see a picture.



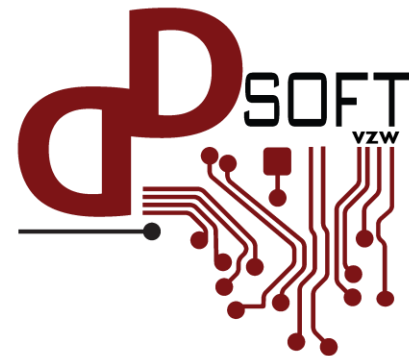
Text To Speech

- ▶ In English for each mobile OS
- ▶ Windows 10 UWP Desktop



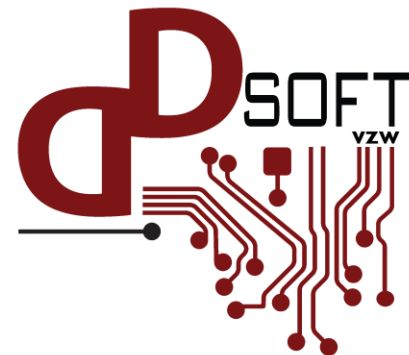
Structured apps

- ▶ Different screens 'views' different content
- ▶ Frameworks aren't helpful for people with disabilities



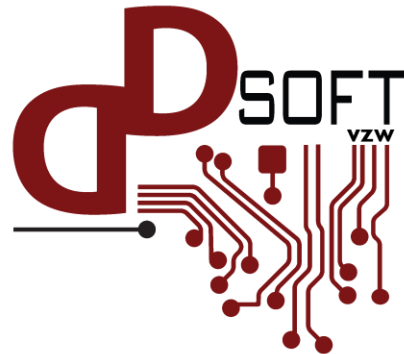
Digital Difficulties for p.w.d.

- ▶ Difficulties are different **par disability**
- ▶ Most helping apps are made **par disability**



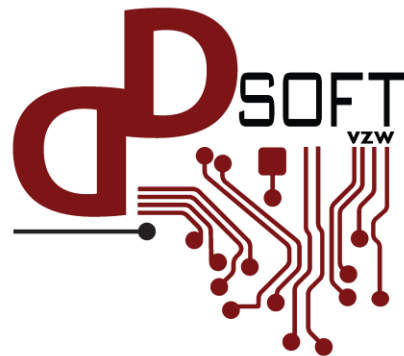
Intellectual Disability

- ▶ Difficulty to:
 - ▶ Read
 - ▶ Understand



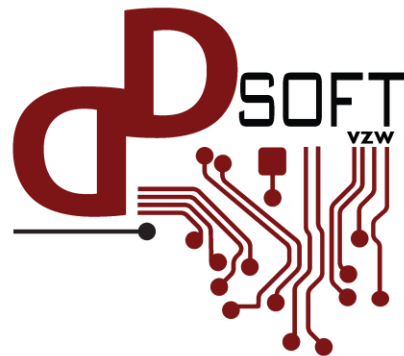
Autism

- ▶ Difficulty to:
 - ▶ Understand
 - ▶ See structure



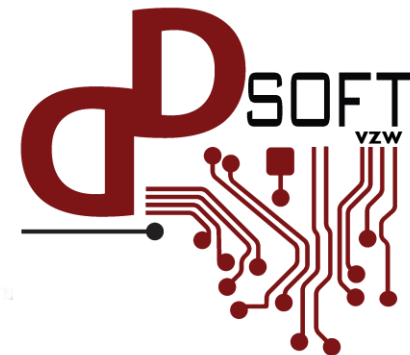
Blindness or vision loss

- ▶ Decreased ability to read
- ▶ Decreased ability to navigate

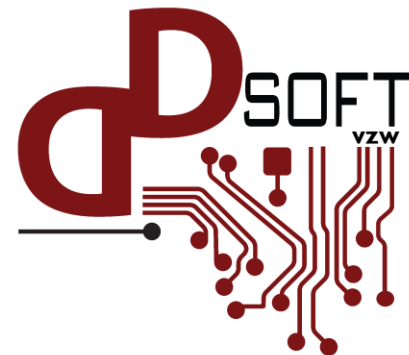


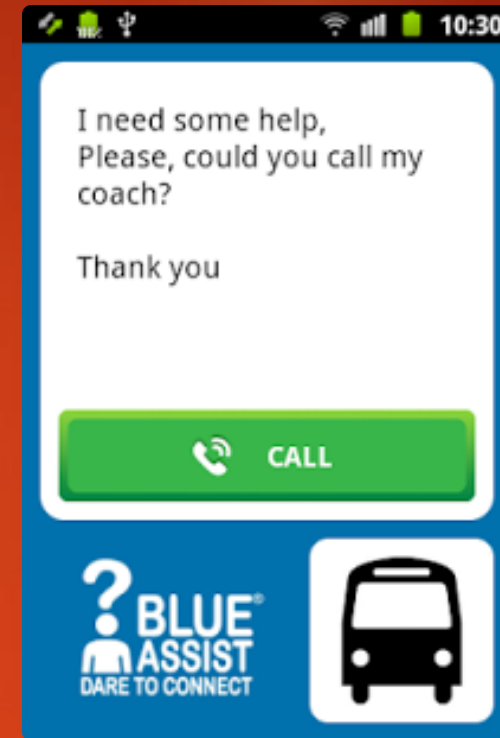
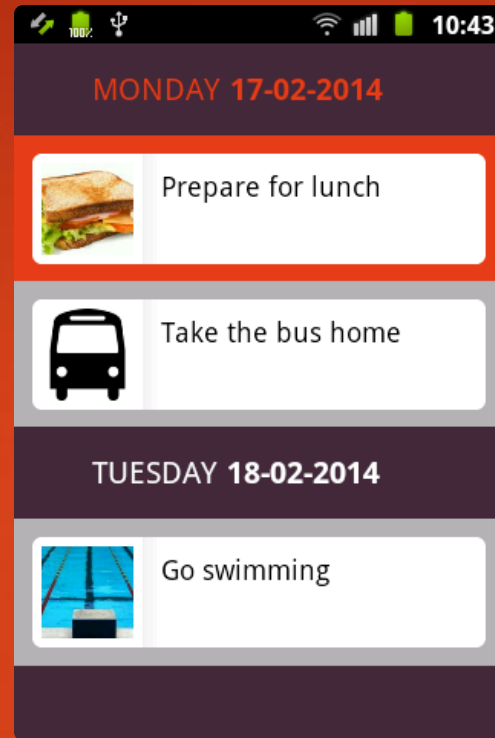
Think inclusive!

- ▶ Do **measure** how many functions needed
- ▶ What information lives with the people.
- ▶ Maybe they want to find their way to the bakery, pizza hut?
- ▶ Apps to **assist** brushing tooth and washing yourself, clothing....
- ▶ Software does matter!
- ▶ Software really helps inclusion.

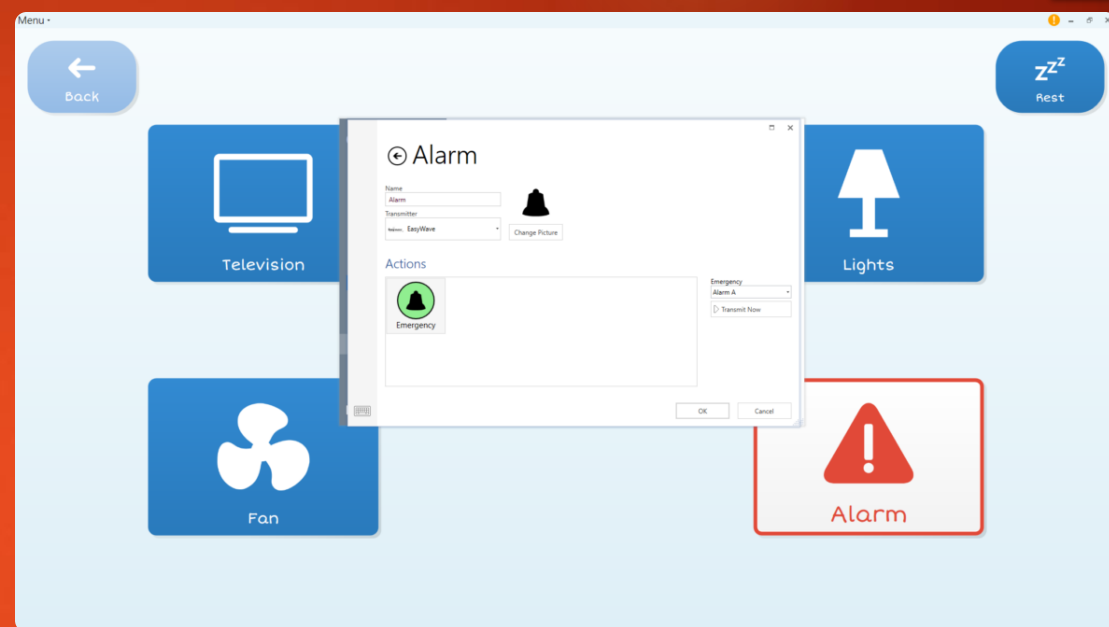
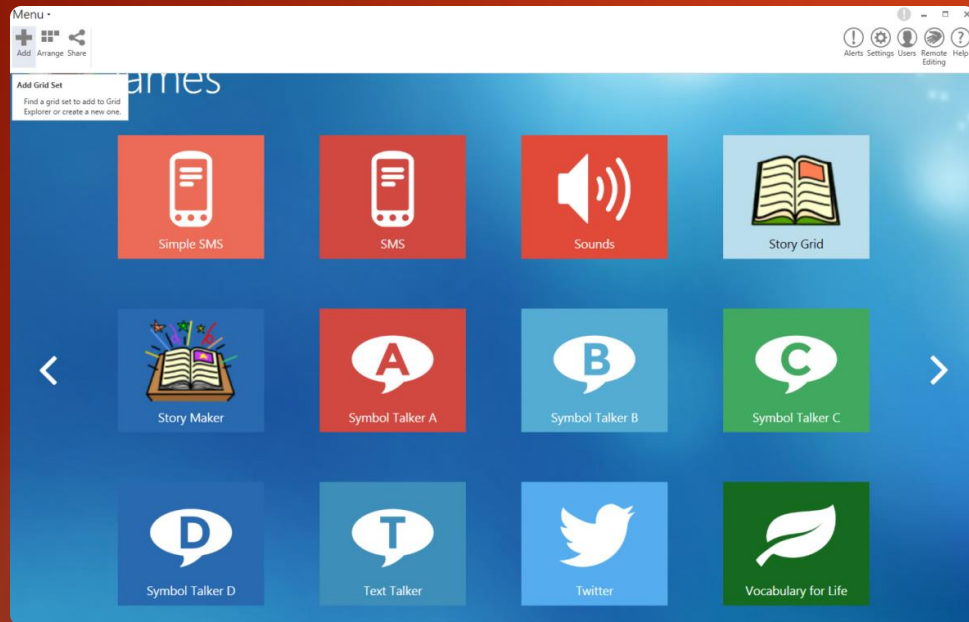


Innovative software examples





Cloudina



Grid 3

Eten (Food)



WOENSDAG 22 MAART 2017



CHAMPIGNONSOEP, STOOFFVLEES MET SLA
EN FRUIT. IFS FRUIT

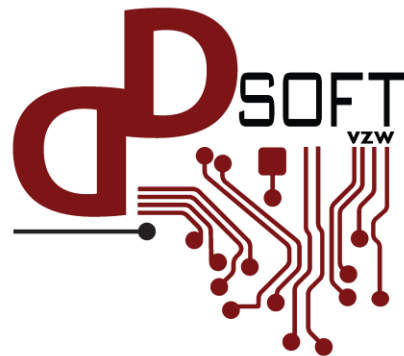


INSTELLINGEN



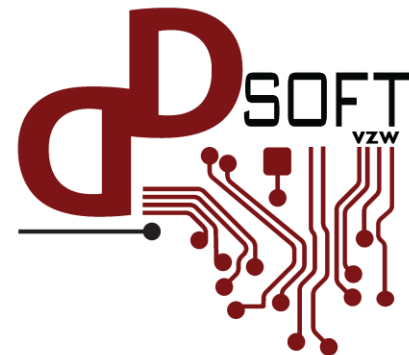
Why does software helps inclusion?

- ▶ Trust
- ▶ ICE
- ▶ Access to info
- ▶ More independency



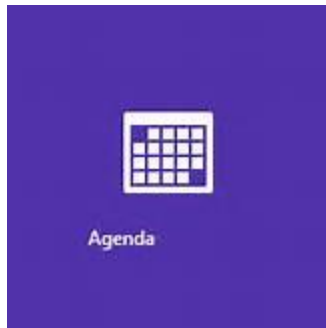
Because a movie tells more than a slide...

► <https://youtu.be/PEi1T8DZHCQ>

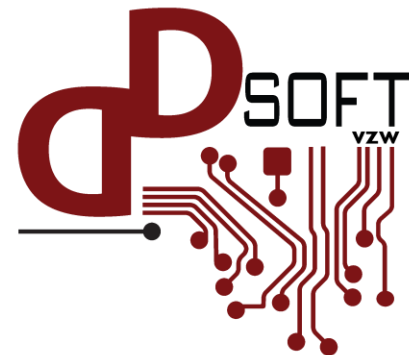


Which apps and software do I use?

- ▶ For me, following apps and software are very helpful

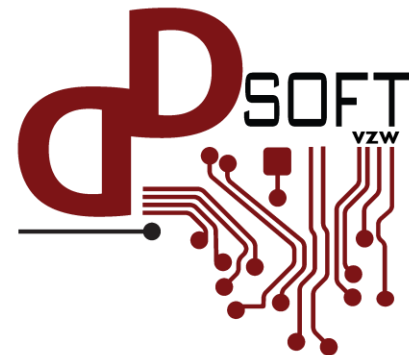


These apps helping me a lot in my daily life!
All of them I use on a daily base



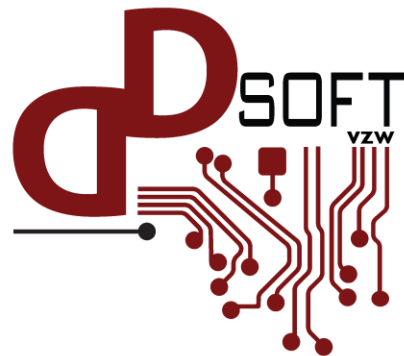
Which apps are missing?

- ▶ Selfcare
- ▶ Wellbeing
- ▶ Navigating
- ▶ Clothing
- ▶ Hygiene
- ▶ Housekeeping
- ▶ Psychological wellbeing



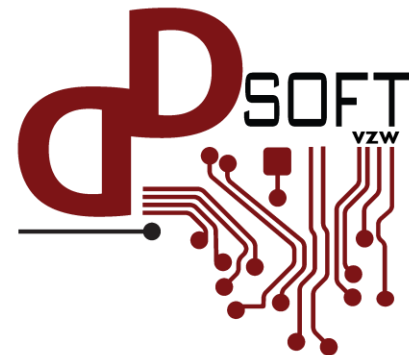
Pricing

- ▶ Less budget
- ▶ Keep it cheap
- ▶ A free side project?



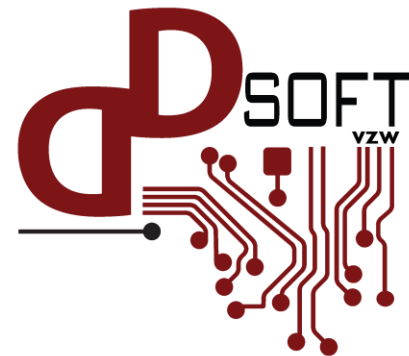
Why should you care?

- ▶ Because, in an inclusive environment where people with a disability takes part of the common life, assistance is needed.
- ▶ As developer YOU:
 - ▶ Can help people find their way.
 - ▶ Can translate info into understandable info for people with a disability.
 - ▶ Can automate progresses.
 - ▶ Can digi-coach.



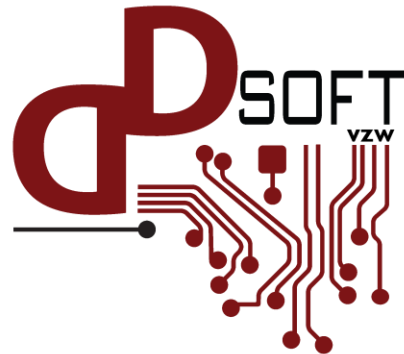
Is there a market?

- ▶ If you mention that more and more people with a disability takes part of the “normal life”, you notice that there will always be more need for assistance.
- ▶ I really think apps can assist!



Does it really helps?

- ▶ Yes! It's proven in test cases!
- ▶ People can find better their road with apps.
- ▶ Can ask better questions with apps.
- ▶ Can get info about their daily life with apps.
- ▶ Can manage their activities with apps.
- ▶ Can do a lot more with apps!



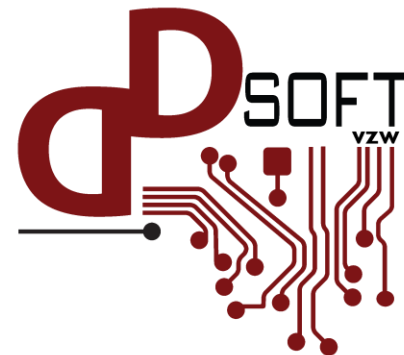
Inclusive software design patterns

Universal Design

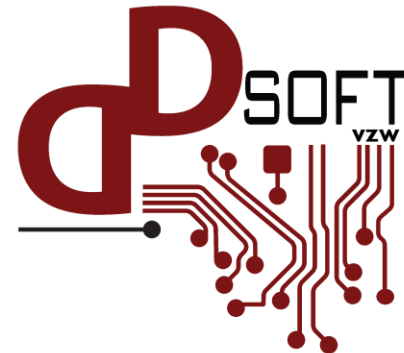
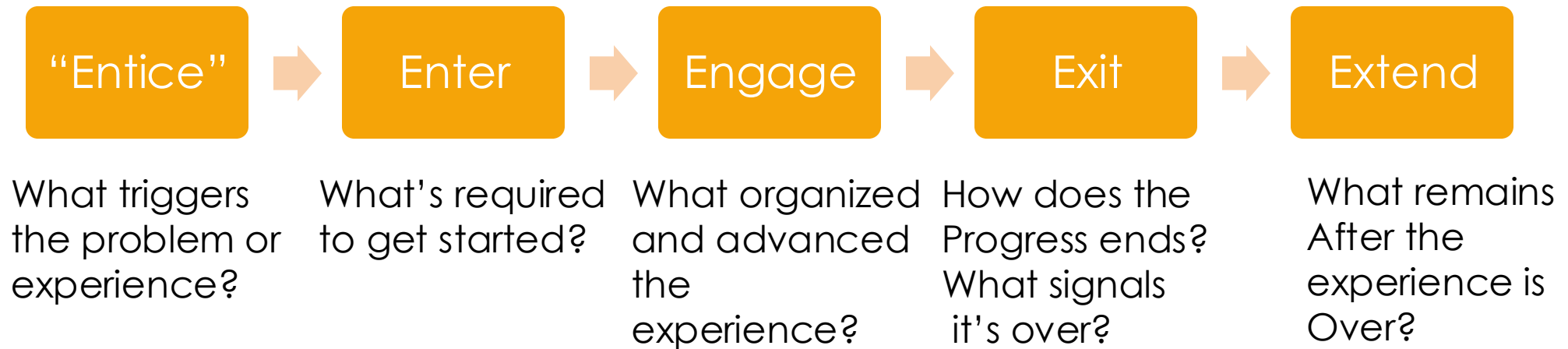
- ▶ Universal Design is the design regardless of age, size, ability or disability.
- ▶ The progress to adopt universal design is as follow:
- ▶ **Discover**: what's the problem?
- ▶ **Define**: which changes are needed?
- ▶ **Develop**: develop a new system / app / software
- ▶ **Deliver**: give the new solution to the people, check if the problem is fixed.

Human Centered Design

- ▶ Human Centered Design is a dynamic way to change the existing world
- ▶ It's about building an ecosystem, setting goals and expectation
- ▶ A good way to work HC is with a 5E Framework.

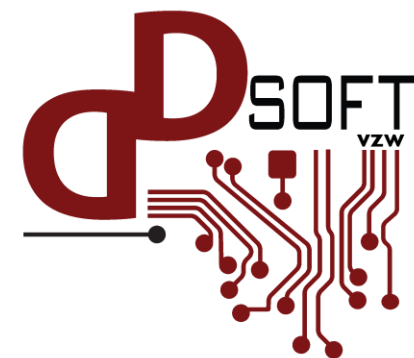


The 5E Framework



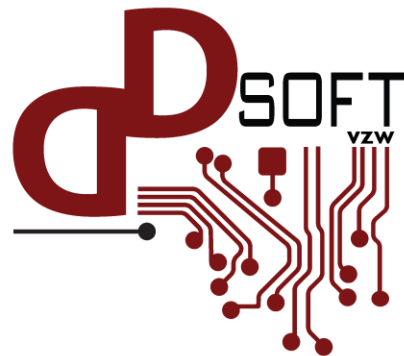
Thank you for listening!

- ▶ Contact:
- ▶ dennie@ddsoft.be
- ▶ +32 478 29 45 25



Bi-directional inclusion

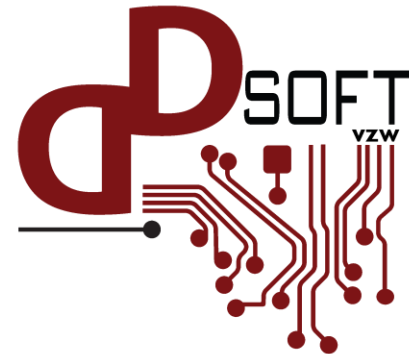
A PEEK AT THE OTHER SIDE



People with autism at work



- ▶ People with autism are mostly:
- ✓ Honest
- ✓ Work good with rules
- ✓ Have a nice planning
- ✓ Are loving structure
- ✓ Direct
- ✓ Very specialized



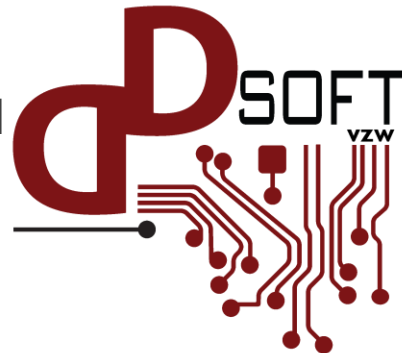
Nothing is such logical as autism.

Developers logic

- A value is true or false
- It's good to declare variables, what type it's expect
- If you program a loop, the exit scenario is mostly clear
- It's very structured

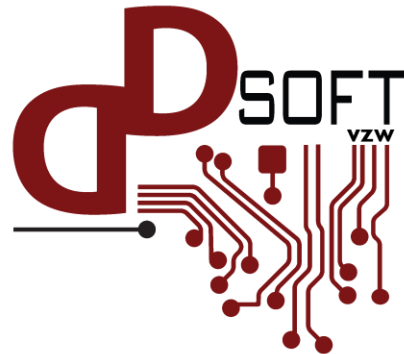
Autistic logic

- ✓ A fact is only true or false
- ✓ It's very nice that every circumstance is declared
- ✓ If there's a path to follow, auti's like it's clear and clean
- ✓ Most of them are very structured



And the last truth:

- ▶ In fact...
- ▶ ... Inclusion is always BI-Directional!!!
- ▶ ... If Else, it is not real inclusion!



Q&A?

